

SOME OUTLIER TESTS FOR MULTIVARIATE SAMPLES

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SUMMARY

Study of multivariate outliers has been sparse and restricted (apart from certain informal descriptive methods) entirely to the normal distribution. This is not unreasonable, in view of the tractability of the multivariate normal distribution. After discussing some general principles in the detection and testing of multivariate outliers the results for the normal case are reviewed. But the main emphasis is on non-normal data. Some *ad hoc* outlier tests are proposed for bivariate uniform, exponential and Pareto distributions and tables of critical values are presented. The results relate principally to fully specified distributions, but some conservative tests are given for cases where parameter values are unknown.

1. Introduction

Model-based multivariate analysis assumes, almost without exception, an underlying multi-normal distribution. The literature is relatively silent on what to do about the analysis of non-normal data. After all, have we not been often reassured that 'the normal distribution is too true not to be good'.

But who has ever seen a multivariate normal sample? Without any conscious bias in their selection a few bivariate samples were chosen and these are shown in Tables 1 and 2, and Figure 1.

It is not claimed that these are typical — but such skewness, and restriction of range, is by no means unusual. If such data are encountered, conventional normal-based methods can hardly be applied resting on the tacit defense for small samples that the data are 'not manifestly non-normal'. The battery of normal-based multivariate methods is of course valuable and inevitable. Rough symmetry about (possibly transformed) co-ordinate axes does appear in multivariate data and provides a pragmatic justification for a normal model, augmented possibly by limit arguments. But the prime stimulus has undoubtedly arisen from the strange tractability of the normal model: a facility of manipulation which is absent when we consider almost any other multivariate data-generating mechanism.

The recent development of non-model-based informal multivariate techniques for data screening, presentation and interpretation provides a welcome release from the commitment to a normal model, but it is high time also to re-exert effort to formulate rich non-normal families of multivariate distributions and to derive the associated statistical methodology.

TABLE 1 Error in sense of perpendicularity v. percentage error in bisection of an angle (1533 observations of females — reduced to age 32.5: from Elderton, Moule and Page, 1928, *Ann. Eugenics*, 3)

		Perpendicularity, Error in Degrees, Central Values											
		0,0625	0,25	0,50	0,75	1,00	1,25	1,50	1,75	2,00	2,25	2,50	Totals
Percentage Error in Bisection, Central Values	0,125	190	54	78,5	21	28	4	4,5	.	2	.	.	382
	0,5	191,5	77	86	23	20	1	6,5	.	2	.	1	408
	1,0	185,5	62	66,5	25	24	2	2	2	1	.	1	371
	1,5	77	24	28	10	13	.	1	.	.	.	.	153
	2,0	50	16	21,5	6	11	2	1	.	.	.	.	107,5
	2,5	22	10	10,5	3	1	.	.	.	1	.	.	47,5
	3,0	15	3	3	3	4	.	.	.	.	.	.	28
	3,5	4	1	3	1	.	.	.	.	.	.	.	9
	4,0	8	2	6	1	1	.	.	.	.	.	.	18
	4,5	.	1	1	1	.	.	.	.	.	.	.	3
	5,0	2	1	.	.	.	.	.	.	.	.	.	3
	5,5	.	.	.	.	.	.	.	.	.	.	.	.
	6,0	1	1	.	1	.	.	.	.	.	.	.	3
Totals		746	252	304	95	102	9	15	2	6	.	2	1 533

TABLE 2 Number of accidents in successive time periods (122 observations; from Arbous & Kerrich, 1951, *Biometrics*, 7, 340-432)

		No. of accidents 1937-1942								
		0	1	2	3	4	5	6	7	Totals
No. of accidents 1943-1947	0	21	18	8	2	1				50
	1	13	14	10	1	4	1			43
	2	4	5	4	2	1	0	1		17
	3	2	1	3	2	0	1			9
	4	0	0	1	1					2
	5		0							0
	6		0							0
	7		1							1
Totals		40	39	26	8	6	2	1	0	122

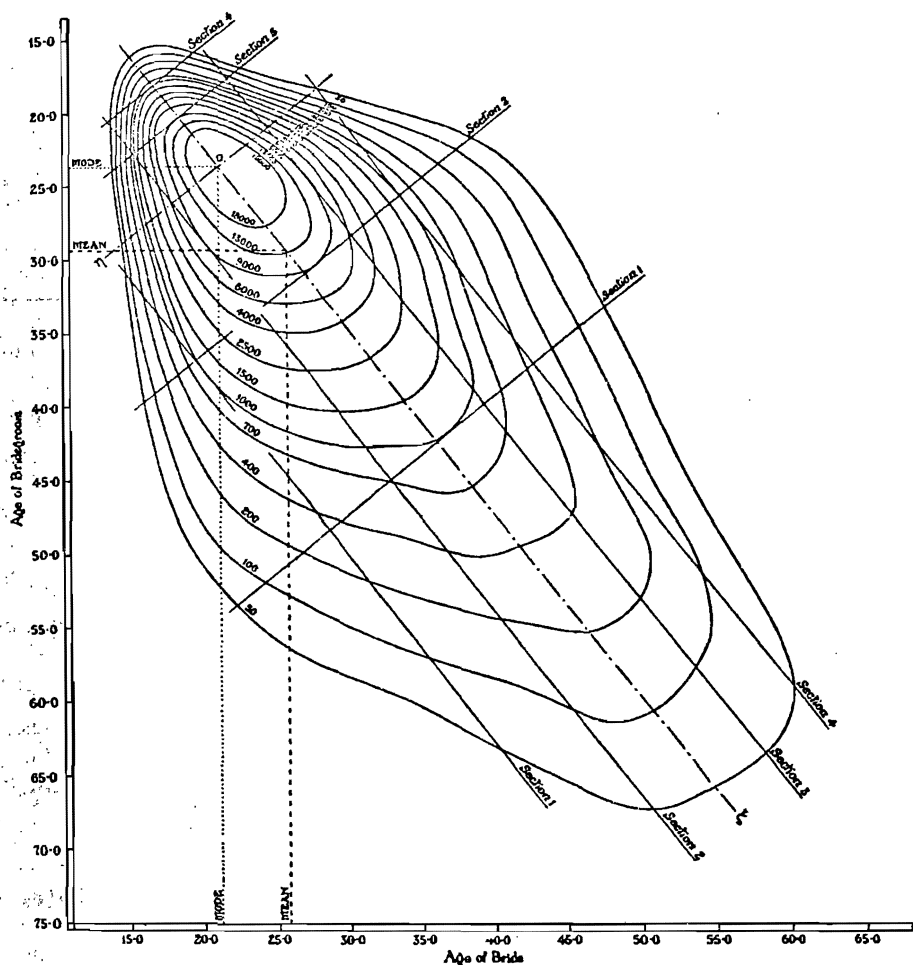


FIGURE 1 Ages of brides and bridegrooms: Australia 1907-1914. (from Pretorius, S.J., 1931, *Biometrika*, 22, 109-223, reproduced by permission of the *Biometrika* trustees).

Such considerations for general multivariate analysis apply even more strongly to the particular problem of studying multivariate outliers. The few results in this area are reviewed in Barnett and Lewis (1978, Ch. 6). Informal procedures, often utilising graphical display, have been developed (e.g. Gnanadesikan and Kettenring, 1972; Hawkins, 1974), but the very nature of an outlier demands at least an implicit model to explain its presence in the data. Model-based study seems here to be restricted entirely to the normal distribution.

This paper makes a modest attempt to awaken interest in the study of outliers in non-normal multivariate samples and provides a few *ad hoc* procedures for testing outliers in specific cases. The detailed results apply to bivariate samples although they often admit ready extension to higher dimensions. In Section 2 some general principles for multivariate outlier modelling are discussed as well as detection and testing and in Section 3 the results for the normal case are reviewed. Sections 4, 5 and 6 are concerned, respectively, with bivariate uniform, exponential and Pareto samples; outlier detection statistics are proposed and associated tests of discordancy developed with corresponding tables of critical values. Seldom do the procedures exploit the general principles outlined in Section 2; more often tractability is achieved by use of intuitively based (inevitably sub-optimal) methods.

2. Detection and testing of multivariate outliers

An outlier in a sample is an 'extreme observation' whose degree of 'extremeness' engenders concern or surprise. Having thus declared an observation to be an outlier; the interest is in testing (by means of a *test of discordancy*) if its 'extremeness' is statistically *unreasonable*. This implies, of course, some null (working) model against which we can assess the degree of 'extremeness'. In developing or contrasting different possible tests of discordancy we will need to formulate also some alternative model which explains the presence of the outlier as the possible manifestation of a discordant value in the sample. (Barnett, 1978b, discusses aims and choice of models in outlier study.)

For univariate samples the notion of 'extremeness' arises from the natural ordering of the observations in the sample — no such facility holds for multivariate samples (Barnett, 1976a) and a test of discordancy must be preceded by (or implicitly include) a mechanism for detecting the outlier. See Barnett and Lewis (1978, Ch. 6) or Barnett (1978a).

Suppose we have a random multivariate sample $\underline{x}_1, \dots, \underline{x}_n$ which we postulate arising from a distribution within a family of distributions $P_{\underline{\theta}}$ indexed by a (possibly vector) parameter $\underline{\theta}$ and having probability (density) $p_{\underline{\theta}}(\underline{x})$. We will need to detect an outlier in terms of the extremeness of the values $R_{\underline{\theta}}(\underline{x}_j)$ ($j=1, 2, \dots, n$) taken by some (usually scalar) metric $R_{\underline{\theta}}(\underline{x})$ over the sample. An example would be the quadratic

distance

$$(\underline{x} - \underline{a})' \underline{\Gamma}^{-1} (\underline{x} - \underline{a})$$

where $\underline{a}$ and $\underline{\Gamma}^{-1}$ measure, respectively, location and dispersion (covariation) in the sample or distribution.

In principle, $R_{\underline{\theta}}(\underline{x})$ might be chosen by the operation of some general statistical criterion; in practice, it is often chosen on the basis of intuitive appeal.

Typically an outlier will be that observation $\underline{x}_i$ which yields the maximum value of $R_{\underline{\theta}}(\underline{x})$ over the sample. If we denote the ordered values of $R_{\underline{\theta}}(\underline{x}_j)$ as $R_{(1)}, \dots, R_{(n)}$ then $\underline{x}_i$ is the observation yielding $R_{(n)}$. Of course, $\underline{\theta}$ may not be known or specified and the outlier will now need to be detected in terms of some associated set of determined values $R_{\underline{\hat{\theta}}}(\underline{x}_j)$ ($j=1, 2, \dots, n$) where $\underline{\hat{\theta}}$ estimates $\underline{\theta}$.

Having detected $\underline{x}_i$ we proceed to test its significance (discordancy) by assessing extremeness of $R_{(n)}$ relative to the working model $P_{\underline{\theta}}$ (with due allowance for estimation of $\underline{\theta}$, if this is necessary). The construction of the test, or at least its statistical properties, will depend on the form of *alternative model* we entertain in place of $P_{\underline{\theta}}$, and various possibilities exist. (Barnett and Lewis, 1978, Ch. 2.)

It could be that the alternative model is of an *inherent* type, replacing $P_{\underline{\theta}}$ by some distinct family $Q_{\underline{\phi}}$ to explain the *complete* sample. Or a *mixture* type alternative model might serve, where $P_{\underline{\theta}}$ is replaced by $(1-\lambda)P_{\underline{\theta}} + \lambda P_{\underline{\phi}}^*$ under which a large proportion $(1-\lambda)$ of observations arise from the original distribution but a small proportion (λ) are alien. $P_{\underline{\phi}}^*$ might, of course, be merely a translated or scaled form of $P_{\underline{\theta}}$. Instead of re-explaining the whole sample, or even just a small proportion of it, the alternative model might merely seek to explain a *single* anomalous member of the sample (or an outlying subsample). A *slippage* type alternative model has this form by declaring that all but one observation (or some prescribed number, k) arise from $P_{\underline{\theta}}$, whilst that one (or subset) arises from $P_{\underline{\phi}}^*$. Again $P_{\underline{\phi}}^*$ might take the form of a simple transformation of $P_{\underline{\theta}}$.

Of course, if $\underline{\theta}$ and $\underline{\phi}$ are (largely) specified we can replace the general families of distributions by sub-families or even prescribed distributions. For the mixture, or slippage, type alternative models to manifest outliers, only particular forms of the component $P_{\underline{\phi}}^*$ will be appropriate. For example, suppose P is a family indexed by a location parameter $\underline{\mu}$ and variance-covariance matrix $\underline{V}$. Then we might have P^* of the same form but with $\underline{\mu}$ replaced by $\underline{\mu} + \underline{a}$ ($\underline{a} \neq \underline{0}$) or with $\underline{V}$ replaced by $b\underline{V}$ ($b \neq 1$). Any $\underline{a}$ might lead

to the manifestation of outlier(s); likewise, any $b > 1$. However, $b < 1$ will not yield 'extreme' observations (seen as outliers); instead it will yield anomalous 'internal' observations of 'non-extreme' form.

In the face of such considerations, how are we to detect outliers and test their discordancy? One possibility is to select a basis for 'ordering' the multivariate sample, declaring an outlier as an extreme value from the chosen standpoint. Some sub-ordering principle of the *reduced* (e.g. based on $R_\theta(\underline{x})$) or *marginal* type (Barnett, 1976a) will need to be used. The null distribution of the associated 'extreme' is then used to assess discordancy. Whilst it would seem that manipulative problems conspire to make such an approach almost inevitable, it is an arbitrary process with no guarantees of statistical propriety or optimality. More formal methods might be tried, but our experience to date is that they are seldom tractable. Such methods include the multiple-decision formulation (used originally by Paulson, 1952, for the univariate normal case) or a Bayesian approach (although no formal test of discordancy is relevant here).

For a slippage-type outlier generating model a well-defined and appealing principle is to use a two-stage maximum likelihood ratio procedure. Denote the working model by H_0 , and the alternative (outlier-generating) model by H_1 and suppose that under H_1 a particular set of $n-k$ observations arise from $P_{\underline{\theta}}$ and the remaining k from $P_{\underline{\phi}}^*$ (with k prescribed). Let the log-likelihoods under H_0 and H_1 be

$$L_0(\underline{x}_1, \dots, \underline{x}_n; \underline{\theta}) \text{ and } L_1[(\underline{x}_{j_1}, \dots, \underline{x}_{j_{n-k}}; \underline{\theta}), (\underline{x}_{j_{n-k+1}}, \dots, \underline{x}_{j_n}; \underline{\phi})],$$

respectively. They are maximised, with values L_0 and $L_1(I_k)$, respectively, when $\underline{\theta}$ and $(\underline{\theta}, \underline{\phi})$ are replaced by the corresponding maximum likelihood estimators (where I_k is that subset of the integers $I = \{1, 2, \dots, n\}$ which indicates the observations provisionally assigned to $P_{\underline{\phi}}^*$). Thus the maximised log-likelihoods differ by an amount

$$L_1(I_k) - L_0$$

under H_1 and H_0 , and it seems sensible to declare as the outlying sub-sample those k observations (if such exist) indicated by I_k^* where

$$0 < L_1(I_k^*) - L_0 = \max_{I_k} (L_1(I_k) - L_0)$$

The outlying sub-sample is then adjudged *discordant* if $L_1(I_k^*) - L_0$ is significantly large under H_0 .

This principle seldom leads to direct explicit detection of the outlier(s) through optimisation of some simple function of the observations. The best prospect is when P_θ and P_ϕ^* are the same family. For example, suppose $\underline{x}_1, \dots, \underline{x}_n$ is, under H_0 , a random sample from $N(\underline{\mu}, \underline{V})$ with $\underline{V}$ known; whilst under H_1 , the observation $\underline{x}_j$ arises from $N(\underline{\mu} + \underline{a}, \underline{V})$, and others from $N(\underline{\mu}, \underline{V})$. Under H_1 we have $\hat{\underline{a}} = \underline{x}_j - \hat{\underline{\mu}}$ and $\underline{x}_j$ makes no contribution to L_1 so that a single outlier is detected as that observation whose omission from the sample most increases the maximised likelihood under H_0 .

A *mixture* or *inherent* type of outlier generating model does not so readily support the two-stage detection (and testing) procedure, although the particular case just discussed suggests a somewhat arbitrary (but pragmatically appealing) alternative. We might detect the outlying sub-sample as that set of k observations whose omission most increases the maximised log-likelihood under H_0 , and adjudge it discordant if the increase is significantly large under H_0 . But this makes no reference to H_1 : our model to explain the presence of outliers.

The two-stage procedure is readily described — it is less easily implemented. Frequently, it yields no *explicit* principle for detecting the outliers and the corresponding null distribution of $L_1(I_k^*) - L_0$ can be prohibitively difficult to handle.

3. The bivariate normal distribution

This is the rare case where the various principles for outlier detection and testing lead to explicit results. For a slippage-type outlier generating model for a single outlier: where P_θ is $N(\underline{\mu}, \underline{V})$ and P_θ^* is $N(\underline{\mu} + \underline{a}, \underline{V})$ it turns out that we want just that observation for which the corresponding (estimated) $R_\theta(\underline{x})$ is a maximum. Specifically, we seek to maximise the estimated distance $R(\underline{x}) = (\underline{x} - \bar{\underline{x}})' \underline{S}^{-1} (\underline{x} - \bar{\underline{x}})$, where $\bar{\underline{x}}$ and $\underline{S}$ are the usual estimators of $\underline{\mu}$ and $\underline{V}$, respectively. Guttman (1973) develops for this model a Bayesian method for the 'detection of spuriousity' which amounts to determining (and acting upon) the posterior distribution of $\underline{a}$. Again the $R(\underline{x})$ metric is encountered.

Also, for a slippage-type model where P_ϕ^* is $N(\underline{\mu}, b\underline{V})$ with $b > 1$, Fergusson (1961) has shown that the above policy has optimal properties when viewed as a multiple-decision procedure. But maximisation of $R(\underline{x})$ is equivalent to minimisation of the *one-outlier scatter ratios*: the intuitively based procedure advanced by Wilks (1963). (See Barnett, 1978a.)

The null distribution of $R_{(n)} = \max_{j=1, \dots, n} R(\underline{x}_j)$ is not very tractable, but reasonable approximations are available and tabulations have been made. (See Wilks, 1963; Barnett and Lewis, 1978). A crucial point is that we can cope with the fact that $\underline{\mu}$ and $\underline{V}$ are unknown – we do not need to tabulate critical values for different values of $\underline{\mu}$ and $\underline{V}$. In the further situations described below this facility does not readily arise, and we need to employ conservative tests or make some assumption (estimate) of the value of $\underline{\theta}$.

4. The bivariate uniform distribution

Perhaps the simplest form of bivariate distribution is that where the component random variables, X_1 and X_2 , are independent and uniformly distributed on known ranges (a,b) , (a',b') . We can for convenience (if necessary using simple linear transformations) assume that X_1 and X_2 have common range $(-d,d)$ centred on the origin. The general methods of Section 3 do not help here, but we can at least set up some pragmatic criteria for outliers.

4.1 Maximum modulus criterion

We might choose to order the observations $(x_{11}, x_{21}) \dots (x_{1n}, x_{2n})$ in terms of the sum of their moduli $|x_{1j}| + |x_{2j}|$, detecting an outlier as that observation yielding the maximum modulus sum. Now $|X_1|$ and $|X_2|$ are independent uniform $(0;d)$ so that $Z = |X_1| + |X_2|$ is triangular on $(0;2d)$. Thus

$$p(Z < z) = 1 - \frac{1}{2}(2 - z/d)^2, (z \geq d),$$

and if $z_j = |x_{1j}| + |x_{2j}|,$

$$(4.1) \quad p\left[\max_{j=1, \dots, n} z_j > \xi\right] = 1 - \left[1 - \frac{1}{2}(2 - \xi/d)^2\right]^n, (\xi \geq d).$$

So the upper α -point of the null distribution for testing discordancy of the outlier (x_{1i}, x_{2i}) yielding $\max_{j=1, \dots, n} z_j$ has the form

$$\xi_\alpha = k_\alpha d$$

where

$$(4.2) \quad k_{\alpha} = \sqrt{2} \{ \sqrt{2} - [1 - (1 - \alpha)^{1/n}]^{\frac{1}{2}} \}, (\alpha > \frac{1}{2}).$$

The k_{α} in (4.2) are readily calculated, and selected 5% and 1% points as shown in Table 3.

TABLE 3 Critical values for maximum modulus test of discordancy for an outlier in a bivariate uniform sample.

n	4	5	6	8	10	12	15	20	30	40	60	100
$\alpha=0.05$	1,840	1,857	1,870	1,887	1,899	1,908	1,917	1,928	1,942	1,949	1,959	1,968
$\alpha=0.01$	1,929	1,937	1,942	1,950	1,955	1,959	1,963	1,968	1,974	1,978	1,982	1,986

4.2 Maximum distance criterion

Alternatively, we might employ scalar distance from the origin to order the observations in the sample. The outlier is then that observation (x_{1i}, x_{2i}) which yields

$$\max_{j=1, \dots, n} w_j \text{ where } w_j = x_{1j}^2 + x_{2j}^2.$$

Now if $U = X_1^2, V = X_2^2$ we have

$$p(U < u) = p(V < u) = \sqrt{u}/d$$

and $F(\psi) = p(W < \psi) = p(U + V < \psi)$ is seen to have the form

$$(4.3) \quad F(\psi) = \begin{cases} \pi\psi/(4d^2), & (\psi \leq d^2) \\ \sqrt{(\frac{\psi}{d^2} - 1)} + \frac{\psi}{2d^2} [\frac{\pi}{2} - 2 \tan^{-1} \sqrt{(\frac{\psi}{d^2} - 1)}], & (\psi > d^2). \end{cases}$$

Thus the upper α -point of the null distribution for testing discordancy of the outlier (x_{1i}, x_{2i}) is η_{α} where $F(\eta_{\alpha}) = (1 - \alpha)^{1/n}$. We find

$$\eta_{\alpha} = \ell_{\alpha} d^2$$

where ℓ_α satisfies

$$(4.4) \quad \ell_\alpha = [\sqrt{(\ell_\alpha - 1)} - (1 - \alpha)^{1/n}] / [\tan^{-1} \sqrt{(\ell_\alpha - 1)} - \pi/4], (\alpha > \frac{1}{2}).$$

Although (4.4) yields no simple explicit form for ℓ_α it is readily solved iteratively to obtain a solution. Table 4 presents some 5% and 1% points.

TABLE 4 Critical values for maximum distance test of discordancy for an outlier in a bivariate uniform sample.

n	4	5	6	8	10	12	15	20	30	40	60	100
$\alpha=0.05$	1.698	1.728	1.751	1.782	1.805	1.821	1.839	1.860	1.885	1.900	1.918	1.937
$\alpha=0.01$	1.862	1.876	1.887	1.901	1.912	1.919	1.928	1.937	1.949	1.956	1.964	1.972

4.3 Outlying areas in relation to rounding errors

One area in which uniform distributions might serve as useful models is in handling rounding or curtailing errors. In some fields of study (notably medicine and industry) there is also the tendency for persons relatively unskilled in taking measurements to exhibit 'digit preference' effects. For example, this is seen in a preponderance (or even a totality) of fives or zeros as the final digits in the recorded numbers. See, for example, Barnett (1976b). The digit preference effect can have serious implications as was shown in the study of cancer diagnoses by liver scan (Barnett, 1978c). Here, doctors were required to take two measurements at right angles to each other on scans and the product of the measurements was used as an assessment of the 'area' of the perfusion defect. An interesting prospect is that of outlying areas, perhaps representing consistent bias in addition to roundoff or digit preference errors. This type of situation prompts the following study.

Suppose that repeated measurements are taken of two of the sides of a square (a rectangle admits obvious modification, or perpendicular diameters on a circle might be measured). The true length of each side is x and we are interested in the area x^2 . The sample measurements are $(x_{11}, x_{21}) (x_{12}, x_{22}) \dots (x_{1n}, x_{2n})$ and we consider a model

$$x_{1j} = x + \epsilon_j$$

$$x_{2j} = x + \eta_j$$

where (ϵ_j, η_j) are independent and uniform on $(-\frac{1}{2}, \frac{1}{2})$. (The measurement scale can always be transformed to accommodate this.) The area errors are

$$(4.5) \quad x_{1j}x_{2j} - \bar{x}^2 = x(\epsilon_j + \eta_j) + \epsilon_j\eta_j$$

Suppose we seek an *upper outlier* among the measured areas. The natural candidate comes from that observation (x_{1j}, x_{2j}) which maximises (4.5), or equivalently $x_{1j}x_{2j}$, and to determine if it is discordant we need the distribution of

$$\max_{j=1, \dots, n} \{(x_{1j}x_{2j})\}.$$

Consider

$$G(\delta) = p(X_1X_2 < \delta) = p(X_1 < \delta/X_2)$$

where $X_1 = x + \epsilon$, $X_2 = x + \eta$ are independent and uniform on $(x - \frac{1}{2}, x + \frac{1}{2})$.

We have

$$p(X_1 < \delta/x_2) = \begin{cases} 0 & , (\delta/x_2 \leq x - \frac{1}{2}) \\ \delta/x_2 - (x - \frac{1}{2}) & , (x - \frac{1}{2} < \delta/x_2 \leq x + \frac{1}{2}) \\ 1 & , (\delta/x_2 > x + \frac{1}{2}). \end{cases}$$

Two cases arise.

I: $\delta > (x - \frac{1}{2})(x + \frac{1}{2})$ where

$$(4.6) \quad G(\delta) = \int_{x - \frac{1}{2}}^{\delta/(x + \frac{1}{2})} dx_2 + \frac{x + \frac{1}{2}}{\delta/(x + \frac{1}{2})} (\delta/x_2 - x + \frac{1}{2}) dx_2$$

$$= \delta \ln \left[\frac{(x + \frac{1}{2})^2}{\delta} \right] + \delta + 1 - (x + \frac{1}{2})^2.$$

II: $\delta \leq (x - \frac{1}{2})(x + \frac{1}{2})$ where

$$(4.7) \quad G(\delta) = \int_{x-\frac{1}{2}}^{\delta/(x-\frac{1}{2})} (\delta/x_2 - x + \frac{1}{2}) dx_2 = \delta \ln \left[\frac{\delta}{(x-\frac{1}{2})^2} \right] - \delta + (x - \frac{1}{2})^2.$$

Clearly, for an upper outlier we will be concerned with the upper part (4.6) of the distribution of $X_1 X_2$. The upper outlier (x_{1j}, x_{2j}) which corresponds with

$\max_{j=1, \dots, n} \{(x_{1j} x_{2j})\}$ will be discordant at level α if $x_{1j} x_{2j}$ exceeds z_α where

$$P \left\{ \max_{j=1, \dots, n} \{(x_{1j} x_{2j})\} > z_\alpha \right\} = \alpha$$

This requires

$$1 - [G(z_\alpha)]^n = \alpha$$

for $G(z_\alpha)$ as given by (4.6). We have

$$z_\alpha = m_\alpha (z + \frac{1}{2})^2$$

where

$$m_\alpha (1 - \ln m_\alpha) = 1 - (1 - \phi)/(x + \frac{1}{2})^2$$

with $\phi = (1 - \alpha)^{1/n}$. Again we have no explicit form for the critical values but they are easily obtained by iterative calculation. Table 5 presents 5% and 1% points for selected sample sizes and a range of values of x . Note a difficulty which is typical of most multivariate outlier problems: namely, that straightforward intuitively based procedures depend on the values of nuisance parameters. Here we need to know x to determine the critical values for a test of discordancy. The sample yields an immediate estimate of x (but in view of the discordant outlier prospect we are best avoiding the otherwise desirable mid-range estimate). We could use the sample estimate of x as a pragmatic basis for assessment of discordancy. A sounder principle, however, would be to try to incorporate the need to estimate x within the construction of the test of discordancy. Although it seems that some success might be achieved here, in general the prospect is not good unless (as

for the multi-normal case) we have a tractable and compact set of minimal sufficient statistics.

TABLE 5 Critical values for a test of discordancy for an upper outlying area (uniformly distributed errors).

n \ x								
		1	2	4	8	12	20	40
4	5%	0,8955	0,9368	0,9647	0,9813	0,9873	0,9922	0,9961
	1%	0,9531	0,9718	0,9843	0,9917	0,9943	0,9965	0,9983
6	5%	0,9143	0,9483	0,9711	0,9847	0,9896	0,9936	0,9968
	1%	0,9617	0,9769	0,9872	0,9932	0,9954	0,9972	0,9986
10	5%	0,9333	0,9598	0,9776	0,9881	0,9919	0,9951	0,9985
	1%	0,9703	0,9821	0,9901	0,9947	0,9964	0,9978	0,9989
15	5%	0,9454	0,9671	0,9817	0,9903	0,9934	0,9960	0,9980
	1%	0,9757	0,9954	0,9919	0,9957	0,9971	0,9982	0,9991
20	5%	0,9527	0,9715	0,9841	0,9916	0,9943	0,9965	0,9982
	1%	0,9789	0,9873	0,9930	0,9963	0,9975	0,9985	0,9992
40	5%	0,9664	0,9798	0,9888	0,9940	0,9960	0,9975	0,9988
	1%	0,9851	0,9910	0,9950	0,9974	0,9982	0,9989	0,9994

5. A Bivariate Exponential Distribution

Bivariate data from distributions with *j*-shaped margins are not uncommon. We have seen examples in Tables 1 and 2. The bivariate Pareto distribution is one such model, admitting positive correlation up to 0,5, and we shall consider it in the next section. For the moment we consider one of the many forms of bivariate exponential distribution (see Johnson and Kotz, 1972, Chapter 41): specifically the one proposed by Gumbel (1960) where (X_1, X_2) has probability density function

$$(5.1) \quad f(x_1, x_2) = e^{-x_1 - x_2 - \theta x_1 x_2} [(1 + \theta x_1)(1 + \theta x_2) - \theta], \quad (x_1, x_2 > 0),$$

with the parameter θ in the range $(0;1)$. X_1 and X_2 are standard exponential marginal distributions and their correlation

$$(5.2) \quad \rho = \theta^{-1} e^{\theta^{-1}} \int_{\theta^{-1}}^{\infty} x^{-1} e^{-x} dt - 1$$

ranges from 0 to -0.4037 as θ goes from 0 to 1.

A typical random sample of 750 observations for the case $\theta=1$ ($\rho=-0.40$) is presented as Figure 2.

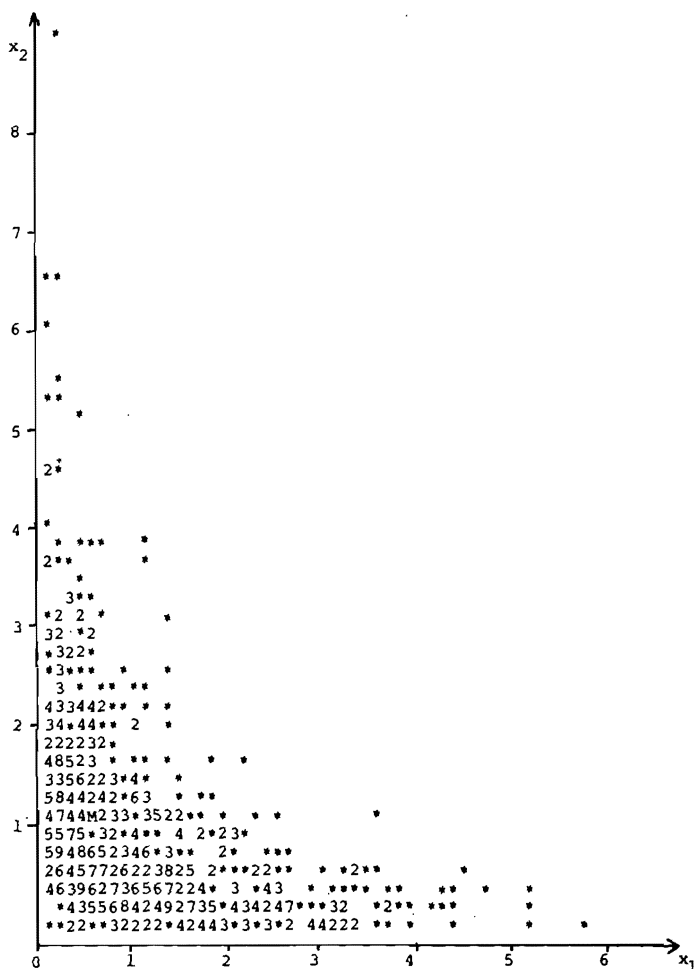


FIGURE 2 Random samples of 750 observations from a bivariate exponential distribution ($\theta=1$; $\rho=-0.4$)

For prescribed θ , and a random sample $(x_{1j}, x_{2j}), (j=1, 2, \dots, n)$ the greatest change in the likelihood (or, more strictly, the joint probability density) arising from dropping one observation occurs for one of the two observations yielding the minimum or maximum values of $x_{1j} + x_{2j} + \theta x_{1j}x_{2j}$. We might declare as the 'upper' outlier, the observation (x_{1i}, x_{2i}) for which $x_{1j} + x_{2j} + \theta x_{1j}x_{2j}$ is a *maximum*. As before we ascribe discordancy to (x_{1i}, x_{2i}) if it is significantly large in relation to its null distribution.

Thus we want the distribution of $U = X_1 + X_2 + \theta X_1 X_2$ and it turns out that the distribution function of $T = 1 + \theta U$ is

$$(5.3) \quad H(t) = 1 - e^{-(t-1)/\theta} [t \ln t / \theta + 1].$$

So the critical value for assessing discordancy at level α is β_α where $H(1 + \theta \beta_\alpha) = (1 - \alpha)^{1/n}$. That is, β_α satisfies

$$(5.4) \quad \beta_\alpha = \ln \left[1 + \frac{(1 + \theta \beta_\alpha) \ln(1 + \theta \beta_\alpha)}{\theta} \right] - \ln [1 - (1 - \alpha)^{1/n}]$$

so that again we must determine the critical values by iterative methods. Table 6 presents 5% and 1% points for a range of sample sizes, and values of θ (with associated ρ , as given by (5.2)).

A special case arises when X_1 and X_2 are independent ($\theta=0$) since $X_1 + X_2$ is now just a $\frac{1}{2} \chi^2_4$ variate and percentage points for the order statistics in a random sample of size n from χ^2_p are tabulated by Gupta (1960) for various values of p . Extracted values relevant to the present problem appear in Barnett and Lewis (1978, table XXVII) where we would need to halve the values tabulated under $p=4$ to cover the present case of $\theta=0$.

The case of prescribed θ is of limited value. When θ is unknown, the problem becomes more difficult. The two-stage maximum likelihood ratio approach proves intractable (and the underlying slippage model is in any case of limited interest). We might just try replacing θ in the above upper outlier criterion by an estimate $\tilde{\theta}$, although the maximum likelihood estimate involves tedious iterative calculation and the distribution of $X_1 + X_2 + \tilde{\theta} X_1 X_2$ appears intractable for any reasonable estimate $\tilde{\theta}$.

A rough and ready approach can be developed using a moment estimator. Suppose we calculate the product moment correlation coefficient r and seek to estimate θ by inverting (5.2) with r replacing ρ . Explicit inversion is impossible, but we can solve for θ approximately by inverting the rational function approximation of Abramowitz and Ste-

TABLE 6 5% and 1% critical values for testing discordancy of an 'upper' outlier in a bivariate exponential sample (θ unknown).

n	α	ρ θ	-0,0844	-0,2773	-0,3835
			0,1	0,5	0,9
4	5%		6,61	7,05	7,25
			8,50	8,97	9,16
	1%				
6	5%		6,87	7,31	7,51
			8,76	9,22	9,42
	1%				
10	5%		7,08	7,53	7,73
			8,97	9,43	9,63
	1%				
15	5%		7,68	8,13	8,33
			9,55	10,02	10,21
	1%				
20	5%		8,15	8,61	8,80
			10,01	10,48	10,67
	1%				
40	5%		8,48	8,94	9,13
			10,33	10,81	10,99
	1%				
40	5%		9,27	9,74	9,93
	1%		11,11	11,59	11,77

gun (1965, p. 231, (5.1.24)) which turns out to produce a maximum error of the order of $2,5 \times 10^{-4}$. All that is involved in obtaining $\tilde{\theta}(r)$ is the solution of a quadratic equation. However, the resulting estimator is far from ideal. Frequently (particularly in small samples) $\tilde{\theta}$ is outside the range (0;1). The bias and mean square error are large. A modified estimator obtained by replacing $\tilde{\theta}$ by 0 or 1 if $\tilde{\theta} < 0$ or $\tilde{\theta} > 1$, respectively, substantially reduces the mean square error, although the bias is still appreciable in small samples.

Using such a modified estimator, $\tilde{\tilde{\theta}}$, we detect an upper outlier as that observation

for which $x_{1j} + x_{2j} + \tilde{\theta} x_{1j}x_{2j}$ is a maximum. However the null distribution does not seem accessible other than by simulation, and large-scale simulations (with numbers of observations of at least 10 000; frequently much higher) have been carried out to obtain approximate 5% and 1% critical values. These are shown in Table 7 for the same ranges of values of θ and n as were used in Table 6. The results are only approximately correct; the second decimal place cannot be guaranteed. Note also that the critical values depend on the true value of θ , which (in small samples at least) is not readily estimable to any

TABLE 7 Approximate 5% and 1% critical values for testing discordancy of an upper outlier in a bivariate exponential sample (θ unknown)

ρ θ n α		$-0,0844$ 0,1	$-0,2773$ 0,5	$-0,3835$ 0,9
4	5%	7,47	7,10	6,69
	1%	10,16	9,24	8,64
5	5%	7,60	7,40	6,96
	1%	10,03	9,49	8,76
6	5%	7,82	7,49	7,18
	1%	10,18	9,50	8,96
10	5%	8,15	8,13	7,79
	1%	10,15	9,96	9,45
15	5%	8,47	8,60	8,29
	1%	10,40	10,38	9,88
20	5%	8,67	8,80	8,67
	1%	10,52	10,57	10,17
40	5%	9,21	9,69	9,47
	1%	10,86	11,31	10,95

great accuracy. Perhaps some consolation arises from the fact that the critical values do not vary too wildly as θ takes different values, and at least a conservative test of discordancy is feasible.

Another prospect is to ignore θ and merely take the upper outlier as that observation (x_{1j}, x_{2j}) for which $x_{1j} + x_{2j} + kx_{1j}x_{2j}$ is a maximum, where the constant k is chosen on some compromise basis from a study of the null distribution of (x_{1j}, x_{2j}) for varying θ . It would be interesting to study this prospect.

6. The Bivariate Pareto Distribution

In contrast to the bivariate exponential distribution (5.1), a bivariate distribution with *positive* correlation is the Pareto distribution of the first kind (Mardia, 1962). Here (X_1, X_2) has joint probability density function

$$(6.1) \quad f(x_1, x_2) = a(a+1)(\theta_1\theta_2)^{a+1}(\theta_2x_1 + \theta_1x_2 - \theta_1\theta_2)^{-(a+2)}, \quad (x_1 > \theta_1 > 0; x_2 > \theta_2 > 0)$$

with $a > 0$. The marginal distributions are univariate Pareto with common shape parameter, but distinct terminals θ_1 and θ_2 . Second order moments exist provided $a > 2$ and the correlation coefficient is

$$(6.2) \quad \rho = a^{-1} \quad (a > 2).$$

Thus $0 < \rho < 0.5$. A random sample of 750 observations when $\rho = 0.4$ is shown in Figure 3, and in contrast with Figure 2 we note how the distinction between negative and positive correlation is somewhat obscured for i-shaped distributions.

Consider first the fully specified case, where a random sample $(x_{11}, x_{21}), \dots, (x_{1n}, x_{2n})$ is available and a, θ_1 and θ_2 are specified. Adopting as the upper outlier the observation whose omission most increases the joint density yields the observation

$$(x_{1j}, x_{2j}) \text{ which maximises } \left[\frac{x_{1j}}{\theta_1} + \frac{x_{2j}}{\theta_2} - 1 \right], \quad (j=1, 2, \dots, n).$$

The distribution of $V = \left[\frac{X_1}{\theta_1} + \frac{X_2}{\theta_2} - 1 \right]$ is readily derived. We find

$$(6.3) \quad p(V > v) = v^{-a}(1 + a - a/v)$$

so that the critical value γ_α for a level- α test of discordancy of the upper outlier is determined from $\{p(V \leq \gamma_\alpha)\}^n = (1-\alpha)$, or specifically

$$(6.4) \quad \delta \gamma_\alpha^{a+1} - (a+1)\gamma_\alpha + a = 0$$

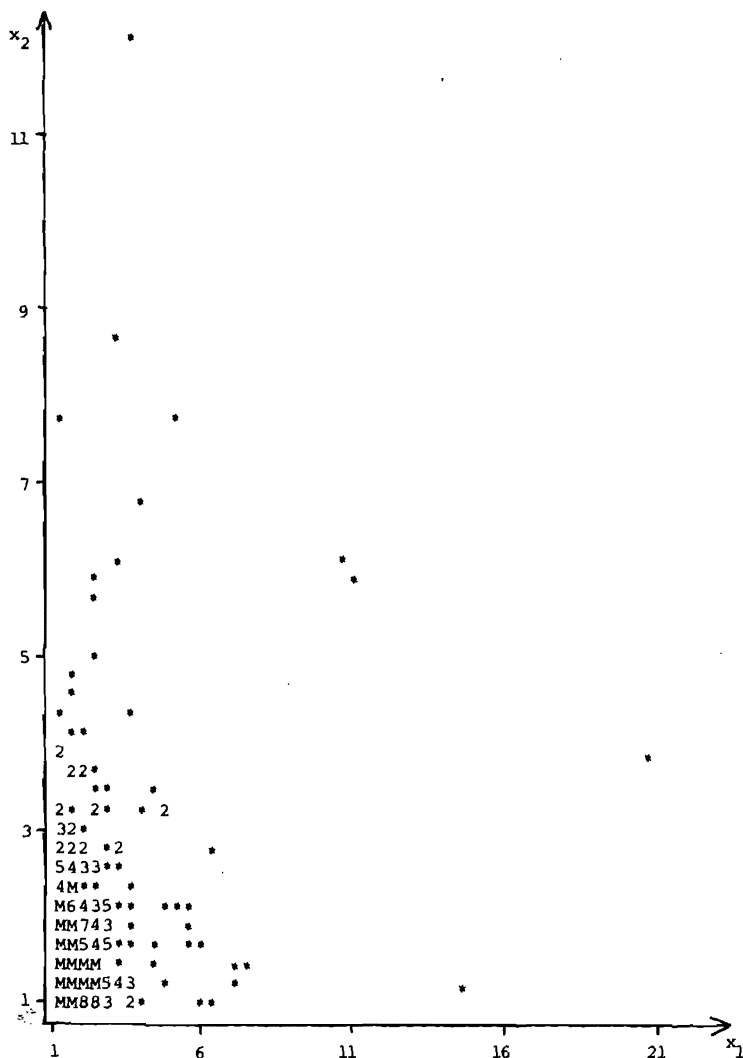


FIGURE 3 Random sample of 750 observations from a bivariate Pareto distribution ($a=2.5$; $\rho=0.4$)

where $\delta = 1 - (1 - \alpha)^{1/n}$.

Once again iterative methods are needed to determine the critical values, and Table 8 presents the 5% and 1% points for a range of values of n and a .

TABLE 8 5% and 1% critical values for testing the discordancy of an upper outlier from a bivariate Pareto distribution (θ_1, θ_2, a known)

n	ρ a	0,4	0,33	0,29	0,25	0,20	0,14	0,10	0,05	0,025
		2,5	3	3,5	4	5	7	10	20	40
4	5%	9,15	6,53	5,10	4,22	3,23	2,35	1,84	1,36	1,17
	1%	17,81	11,42	8,26	6,46	4,55	3,01	2,19	1,49	1,22
5	5%	10,03	7,05	5,45	4,48	3,38	2,43	1,88	1,38	1,18
	1%	19,50	12,32	8,82	6,85	4,76	3,12	2,25	1,51	1,23
6	5%	10,81	7,51	5,75	4,70	3,51	2,50	1,92	1,40	1,18
	1%	21,00	13,11	9,31	7,18	4,95	3,21	2,29	1,53	1,24
8	5%	12,16	8,29	6,27	5,07	3,74	2,61	1,98	1,42	1,19
	1%	23,59	14,46	10,13	7,73	5,26	3,35	2,36	1,55	1,25
10	5%	13,32	8,95	6,70	5,37	3,92	2,70	2,03	1,44	1,20
	1%	25,82	15,59	10,81	8,19	5,51	3,46	2,42	1,57	1,26
12	5%	14,35	9,52	7,06	5,63	4,07	2,78	2,07	1,45	1,21
	1%	27,80	16,58	11,40	8,58	5,72	3,56	2,47	1,58	1,26
15	5%	15,71	10,28	7,55	5,97	4,27	2,88	2,12	1,47	1,21
	1%	30,42	17,89	12,17	9,08	5,99	3,68	2,52	1,60	1,27
20	5%	17,66	11,34	8,22	6,43	4,53	3,01	2,19	1,49	1,22
	1%	34,16	19,71	13,23	9,77	6,38	3,84	2,60	1,63	1,28
30	5%	20,82	13,02	9,26	7,14	4,93	3,20	2,28	1,52	1,24
	1%	40,23	22,60	14,89	10,84	6,91	4,08	2,72	1,67	1,29
40	5%	23,40	14,36	10,07	7,69	5,24	3,34	2,36	1,55	1,25
	1%	45,17	24,90	16,18	11,67	7,33	4,26	2,80	1,69	1,30

The more interesting case is where the parameters are unknown. As a start suppose θ_1 and θ_2 are unknown, but a is known. The two-stage likelihood ratio approach, with a slippage alternative model under which θ_1 and θ_2 are larger for the anomolous observation, looks promising as a means of examining an upper outlier. In Mardia (1964), we see that the maximum likelihood estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ of θ_1 and θ_2 are the marginal minima $x_{1(1)}$ and $x_{2(1)}$, respectively. The two stage procedure now detects as upper outlier the observation (x'_{1i}, x'_{2i}) yielding

$$\max_{j=1, \dots, n} \left[\frac{x_{1j}}{x_{1(1)}} + \frac{x_{2j}}{x_{2(1)}} - 1 \right] \text{ provided neither } x_{1i} = x_{1(1)} \text{ or } x_{2i} = x_{2(1)}.$$

But it is unlikely that the upper outlier would have one or other of its components as a marginal minimum, especially in samples of any reasonable size. Thus (x'_{1i}, x'_{2i}) seems a reasonable pragmatic choice. Unfortunately, however, $\hat{\theta}_1$ and $\hat{\theta}_2$ do not have quite the simple form described originally by Mardia (1964). In recent correspondence, he has indicated that it is only for particular configurations of sample that $\hat{\theta}_1 = x_{1(1)}$, $\hat{\theta}_2 = x_{2(1)}$; otherwise $\hat{\theta}_1$ or $\hat{\theta}_2$ will only be defined implicitly through relationships of the form

$$(6.5) \quad \left\{ \begin{array}{l} \sum_{j=1}^n \left\{ \frac{x_{\ell j}}{\hat{\theta}_{\ell}} \left[\frac{x_{\ell j}}{\hat{\theta}_{\ell}} + \frac{x_{kj}}{x_{k(1)}} - 1 \right] \right\} = k(x) \\ \hat{\theta}_k = x_{k(1)} \end{array} \right.$$

where $(\ell; k)$ is $(1; 2)$ or $(2; 1)$.

Notwithstanding this limitation on the utility of $x_{1(1)}$ and $x_{2(1)}$ for estimating $\hat{\theta}_1$ and $\hat{\theta}_2$, the determination of the upper outlier from maximisation of

$$\frac{x_{1j}}{x_{1(1)}} + \frac{x_{2j}}{x_{2(1)}} - 1$$

is still intuitively appealing. But what can we say of the null distribution of the resulting (x_{1i}, x_{2i}) ? I have not succeeded in deriving the exact distribution although a conservative test is easily recognised, and the true critical values have been estimated by simulation.

Note first of all that

$$\frac{x_{1j}}{x_{1(1)}} + \frac{x_{2j}}{x_{2(1)}} - 1 \leq \frac{x_{1j}}{\theta_1} + \frac{x_{2j}}{\theta_2} - 1$$

for each $j=1, 2, \dots, n$. Thus

$$\max_{j=1, \dots, n} \left\{ \frac{x_{1j}}{x_{1(1)}} + \frac{x_{2j}}{x_{2(1)}} - 1 \right\} \leq \max_{j=1, \dots, n} \left\{ \frac{x_{1j}}{\theta_1} + \frac{x_{2j}}{\theta_2} - 1 \right\},$$

and the critical values for a test based on (x_{1j}, x_{2j}) must be greater than those for a test based on (x'_{1j}, x'_{2j}) . So for the case when θ_1 and θ_2 are unknown, the earlier derived

TABLE 9 5% and 1% critical values for testing the discordancy of an upper outlier from a bivariate Pareto distribution (a known)

n	α	ρ a	0,4	0,33	0,29	0,25	0,20	0,14	0,10	0,05	0,025
			2,5	3	3,5	4	5	7	10	20	40
4	5%		8,11	5,89	4,62	3,92	3,02	2,25	1,75	1,34	1,15
	1%		16,16	9,95	7,67	5,83	4,10	2,86	2,11	1,46	1,20
5	5%		8,93	6,61	5,12	4,22	3,22	2,31	1,82	1,35	1,17
	1%		17,26	11,31	8,33	6,36	4,54	3,04	2,17	1,47	1,22
6	5%		9,98	6,98	5,41	4,49	3,40	2,42	1,85	1,38	1,17
	1%		20,37	11,91	8,55	6,91	4,70	3,16	2,20	1,51	1,23
8	5%		11,58	7,91	5,99	4,83	3,64	2,53	1,95	1,40	1,19
	1%		21,15	12,63	9,64	7,40	5,21	3,32	2,34	1,53	1,23
10	5%		12,67	8,76	6,51	5,21	3,88	2,68	2,01	1,43	1,20
	1%		24,30	15,65	10,70	7,95	5,56	3,37	2,42	1,57	1,25
12	5%		13,93	9,43	7,02	5,54	4,05	2,71	2,04	1,44	1,20
	1%		26,71	16,64	11,20	8,62	5,81	3,56	2,41	1,57	1,26
15	5%		15,16	9,86	7,40	5,90	4,21	2,81	2,11	1,46	1,21
	1%		29,31	17,18	12,13	9,02	5,82	3,57	2,52	1,59	1,26
20	5%		16,83	11,14	8,02	6,22	4,50	2,93	2,19	1,47	1,22
	1%		33,80	18,79	13,07	9,40	6,26	3,77	2,58	1,63	1,28
30	5%		20,39	12,69	9,10	6,96	4,89	3,17	2,27	1,52	1,23
	1%		40,72	22,00	14,73	10,59	6,95	4,06	2,78	1,66	1,30
40	5%		23,10	14,48	10,16	7,80	5,18	3,34	2,36	1,54	1,25
	1%		45,05	26,24	16,18	12,09	7,21	4,16	2,80	1,69	1,30

values γ_α (see 6.4) provide upper bounds for the upper α -points and the entries in Table 8 provide the basis for a conservative test. Just how conservative this test is, has been examined by estimating the upper 5% and 1% points from large-scale simulation and the results are presented in Table 9. Throughout most of the table, estimates derive from samples of size 10 000; in some cases the sample sizes were 20 000. Some indication of inaccuracy is apparent from a few of the entries in Table 9 (for large n and small α , and predominantly at the 1% levels) slightly exceeding the corresponding values in Table 8. Typically, the accuracy cannot be guaranteed in the third significant figure for small α (especially when n is large). The vast effort involved in estimating upper percentage points has prohibited greater accuracy at this stage. The further case when α is unknown requires separate investigation.

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